

UNITED STATES PATENT APPLICATION

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COUNTRY OF RESIDENCE: United States of America

ATTORNEY DOCKET NO.: BLM-006

TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to autonomous content performance optimization systems, and more particularly to a HyperLearning engine that combines a 72x human-equivalent learning acceleration multiplier, simultaneous multi-variate A/B testing across 30 parallel test variants, detection of 15-dimensional micro-patterns in content performance data, cross-platform behavioral synthesis, composite predictive modeling, and deadline pressure response to maximize content engagement across social media platforms.

BACKGROUND OF THE INVENTION

Human social media managers analyze content performance through a limited number of dimensions (typically 5 or fewer: engagement rate, reach, impressions, click-through, and saves) and make optimization decisions on a weekly or monthly cadence. This slow, low-dimensional analysis cycle means that content performance patterns are discovered months after they first emerge and optimization decisions are made on stale data.

Automated content optimization systems improve on the cadence but typically operate on the same low-dimensional feature space as human analysts, and do not operate at the statistical precision needed to detect micro-patterns – subtle correlations between specific content formatting choices (e.g., the presence of a question mark in the second sentence, the ratio of capital letters to total characters, the position of a URL relative to hashtags) and engagement outcomes. These micro-patterns are invisible to human analysis because they require hundreds of data points per pattern to achieve statistical significance, and they are platform-specific: a micro-pattern that improves engagement on TikTok may have no effect on LinkedIn.

Furthermore, existing systems do not synthesize micro-patterns across platforms to identify universal patterns (effective on

all platforms) versus platform-specific amplifiers (effective only on specific platforms), limiting the ability to generate content that performs well everywhere simultaneously.

There exists a need for a content optimization engine that: (1) applies a human-equivalent learning acceleration multiplier to perform analysis cycles in minutes that a human analyst would take months to complete; (2) simultaneously tests 30 content variants per cycle; (3) detects micro-patterns across 15 formatting dimensions with statistical significance testing; (4) synthesizes platform-specific and universal patterns into a unified cross-platform content model; and (5) responds to deadline pressure by adjusting the learning rate and analysis focus.

SUMMARY OF THE INVENTION

The present invention provides a HyperLearning engine comprising a 72x human-equivalent learning acceleration multiplier, a 30-variate simultaneous A/B testing framework, a 15-dimensional micro-pattern detection system, a cross-platform behavioral synthesis layer, a composite predictive model combining timing, content, hashtag, hook, and format predictions, and a deadline pressure response mechanism.

In a first aspect, the invention provides a learning acceleration framework that defines a base human analyst learning rate of 1.0, an owner learning rate of 24x faster than a human analyst, an owner capacity multiplier of 3x, producing a compound learning multiplier of 72x ($= 24 \times 3$), enabling the engine to perform in 5 minutes what a human analyst would take 6 hours to complete.

In a second aspect, the invention provides a 30-variate simultaneous A/B testing system that requires a minimum of 30 impressions per variate before statistical testing, applies an 80% significance threshold for declaring a winner, and extracts

learnings from every test regardless of winner status.

In a third aspect, the invention provides a micro-pattern detector that analyzes 15 structural dimensions of content: character count, emoji density, hashtag position, word sentiment, timing precision, media aspect ratio, color temperature, call-to-action placement, hook structure, line breaks, question marks, exclamation density, capital letter ratio, number usage, and URL position. Each micro-pattern is characterized by type, pattern description, correlation coefficient, sample size, confidence interval, engagement impact, and a list of platforms where the pattern is significant.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. Learning Acceleration Framework

The HyperLearning engine defines three constants:

HUMAN_BASELINE = 1.0 (reference learning rate: one human analyst)

OWNER_LEARNING_RATE = 24.0 (24× faster analysis cycle than human)

OWNER_MULTIPLIER = 3.0 (3× capacity: simultaneous analysis of 3 workstreams)

LEARNING_MULTIPLIER = 72.0 (= 24 × 3: compound acceleration factor)

The analysis cycle runs every 5 minutes (matching the process-level cache TTL). At 72× acceleration, a 5-minute cycle is equivalent to 360 minutes (6 hours) of human analyst work. The engine analyzes $5 \times \text{LEARNING_MULTIPLIER} = 360$ analysis dimensions simultaneously (versus the 5 dimensions a human analyst can hold in working memory).

HUMAN_ANALYSIS_DIMENSIONS = 5

$$\text{HYPER_ANALYSIS_DIMENSIONS} = 5 \times 72 = 360$$

II. 30-Variate Simultaneous A/B Testing

The HyperLearning engine runs 30 simultaneous content variants (HYPER_AB_VARIATES = 30) per test cycle. Each variate is a distinct combination of content format, hook structure, hashtag set, timing window, and platform parameters. Variates are tracked until each accumulates at least 30 impressions (AB_MIN_IMPRESSIONS_PER_VARIATE = 30), at which point statistical significance testing is applied.

A winner is declared when a variate's engagement rate is statistically significantly better than all others at the 80% confidence threshold (AB_SIGNIFICANCE_THRESHOLD = 0.80). Learnings are extracted from every test including inconclusive tests, because knowing that a particular content format has no measurable effect on engagement is itself actionable information.

III. 15-Dimensional Micro-Pattern Detection

The micro-pattern detector analyzes each content string and its associated engagement outcome across 15 structural dimensions:

1. character_count: total character length
2. emoji_density: emoji count / total word count
3. hashtag_position: position of first hashtag (start, middle, end)
4. word_sentiment: average valence of content words
5. timing_precision: granularity of posting time (hour, half-hour, quarter-hour)
6. media_aspect: aspect ratio of attached media (square, portrait, landscape)
7. color_temperature: dominant color temperature of attached media (warm/cool/neutral)

8. cta_placement: position of call-to-action relative to total content length
9. hook_structure: grammatical structure of opening sentence (question/statement/command)
10. line_breaks: number of paragraph breaks in content
11. question_marks: count of question marks
12. exclamation_density: exclamation marks / total sentences
13. capital_ratio: uppercase characters / total alphabetic characters
14. number_usage: count of numeric tokens in content
15. url_position: position of URL relative to hashtags (before/after/absent)

For each dimension, the detector computes a Pearson correlation coefficient between the dimension value and the observed engagement rate across a sample of published posts. A micro-pattern is recorded when the sample size meets a minimum threshold and the correlation magnitude exceeds a significance threshold. Each micro-pattern entry includes: id, type, pattern description, correlation, sample_size, confidence, engagement_impact, platform_specific flag, and a list of platforms where the pattern has been confirmed.

IV. Cross-Platform Behavioral Synthesis

The cross-platform synthesis layer aggregates micro-patterns across platforms to produce:

- A. Universal patterns: micro-patterns confirmed on all or most platforms with consistent direction (e.g., "question hook increases engagement by +0.8% across all platforms")
- B. Platform-specific amplifiers: micro-patterns that are significant on one platform but not others (e.g., "emoji density > 0.2 increases TikTok engagement by +2.1% but has no effect on LinkedIn")

C. Optimal content matrix: a multi-dimensional matrix of content feature combinations with predicted engagement scores and confidence intervals for each combination, organized by platform

D. Audience behavior model: captures peak activity windows, content fatigue cycles (the period after high-volume posting during which engagement drops), engagement velocity curves (engagement rate as a function of hours since posting), and virality threshold estimates per platform

V. Composite Predictive Model

The HyperLearning engine maintains five predictive sub-models:

timing: predicts optimal posting hour and day-of-week per platform from historical engagement data

content: predicts engagement rate from content feature vector

hashtag: predicts engagement impact of specific hashtag combinations

hook: predicts click-through rate from hook structure and opening word choices

format: predicts platform algorithm favorability from format parameters (length, media type, aspect)

composite: weighted combination of all five sub-models

Each prediction includes a predicted engagement value, confidence interval, and a list of contributing factors with their individual weights. The composite model is used by the content generation system to construct optimal content recommendations.

VI. Process-Level Query Cache

To avoid executing multiple expensive database aggregate queries within a single 5-minute analysis cycle, the engine maintains a process-level in-memory cache keyed by query

identifier with a 6-minute TTL. Each distinct aggregate query executes against the database at most once per cycle window, with subsequent calls within the window reading from the cache. This eliminates all within-cycle redundant I/O without requiring any external cache infrastructure.

VII. Deadline Pressure Response

The engine exposes an `applyDeadlinePressure(pressure: number)` method that adjusts the learning behavior under schedule pressure:

- pressure = 0: normal operation – balanced exploration of all 15 micro-pattern dimensions

- pressure > 0.5: accelerated mode – focus on the 5 highest-correlation micro-patterns

- pressure > 1.5: critical mode – exploit only the single highest-engagement pattern per platform

Under critical pressure, the engine's optimal content recommendation shifts from the balanced composite prediction to the highest-yield single-variate recommendation for the target platform, trading prediction accuracy for generation speed.

VIII. Optimal Content Prediction Output

The engine exposes a `predictOptimalContent(userId, platform)` method that returns:

- optimalHook: recommended opening sentence structure

- optimalLength: recommended total character count range

- optimalTiming: recommended posting hour and day-of-week

- microPatternRecommendations: ordered list of actionable micro-pattern improvements

- predictedEngagement: predicted engagement rate for the recommended combination

These outputs are consumed by the autonomous content generation system to apply micro-pattern optimizations to generated content before the quality gate evaluation.

IX. Hyper A/B Micro-Pattern Application

Before publishing, the content generation system queries the engine for micro-pattern recommendations and applies up to three of them to the generated content:

If emoji recommendation applies and content has no emoji:
prepend a domain-appropriate emoji

If question recommendation applies and content has no
question: append "What do you think?"

Additional micro-pattern applications follow the same
additive pattern

This application step is non-destructive: it adds micro-pattern improvements without replacing the generated content, preserving the quality gate score while adding engagement-boosting formatting signals.

CLAIMS

1. A computer-implemented content optimization engine, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: define a learning acceleration multiplier as a product of a per-operator learning rate advantage over a human baseline analyst and a per-operator capacity multiplier; run a plurality of simultaneous content variant tests per analysis cycle, requiring a minimum impression count per variant before applying statistical significance testing; detect correlations between content structural dimensions and engagement outcomes across a plurality of structural dimensions exceeding the number of dimensions a human analyst can simultaneously track; and synthesize detected micro-patterns across platforms into universal patterns applicable across all platforms and platform-specific amplifiers applicable to individual platforms.
2. The system of claim 1, wherein the learning acceleration multiplier is 72, derived from a per-operator learning rate of 24 times the human baseline multiplied by a capacity multiplier of 3, enabling the engine to complete an analysis cycle equivalent to 6 hours of human analyst work within 5 real minutes.
3. The system of claim 1, wherein the plurality of content variant tests is 30 simultaneous variants per cycle, requiring a minimum of 30 impressions per variant and declaring a winner when a variant's engagement rate exceeds all others at a significance threshold of at least 80 percent.
4. The system of claim 1, wherein the plurality of structural dimensions comprises at least 15 dimensions including

character count, emoji density, hashtag position, word sentiment, timing precision, media aspect ratio, color temperature, call-to-action placement, hook structure, line break count, question mark count, exclamation density, capital letter ratio, numeric token count, and URL position relative to hashtags.

5. The system of claim 1, wherein each detected micro-pattern is characterized by a type identifier, pattern description, Pearson correlation coefficient, sample size, confidence interval, engagement impact value, platform-specificity flag, and a list of platforms on which the pattern has been confirmed.
6. The system of claim 1, further comprising a cross-platform synthesis layer that combines platform-specific micro-patterns into: a set of universal patterns confirmed effective across all platforms; a map of platform-specific amplifier patterns; an optimal content feature matrix with predicted engagement scores by feature combination; and an audience behavior model comprising peak activity windows, content fatigue cycles, engagement velocity curves, and virality thresholds.
7. The system of claim 1, further comprising five predictive sub-models for timing, content features, hashtag combinations, hook structure, and content format, combined in a composite predictive model that returns a predicted engagement rate, confidence interval, and a list of contributing factors with individual weights.
8. The system of claim 1, further comprising a process-level in-memory query cache with a configurable time-to-live that stores results of database aggregate queries for the duration of one analysis cycle, ensuring each distinct query executes against the database at most once per

cycle.

9. The system of claim 1, further comprising a deadline pressure response mechanism that narrows the set of micro-patterns applied under increasing schedule pressure, applying all micro-patterns under no pressure, focusing on highest-correlation patterns under moderate pressure, and applying only the single highest-yield recommendation under critical pressure.
10. The system of claim 1, further comprising a micro-pattern application layer that, given a set of micro-pattern recommendations and a generated content string, applies up to a configurable number of non-destructive micro-pattern improvements to the content without altering content generated by the primary generation engine.
11. The system of claim 10, wherein the non-destructive micro-pattern improvements include prepending a domain-appropriate emoji when the emoji density micro-pattern recommends higher emoji density and the content contains no emoji, and appending an engagement prompt when the question mark micro-pattern recommends increased question density and the content contains no question mark.
12. A method of detecting and applying content micro-patterns for engagement optimization, the method comprising: collecting a dataset of content strings paired with observed engagement rates; computing a correlation coefficient between each of a plurality of structural content dimensions and the observed engagement rates; recording each dimension whose correlation magnitude exceeds a significance threshold at a minimum sample size as an active micro-pattern with its correlation, confidence, and engagement impact; classifying

each active micro-pattern as universal if confirmed across all platforms or platform-specific if confirmed on a proper subset of platforms; and applying the top-ranked active micro-patterns to newly generated content as non-destructive formatting improvements.

13. The method of claim 12, wherein collecting a dataset comprises aggregating engagement performance records from a plurality of social media platforms and normalizing engagement rates by platform-specific baseline engagement rates to enable cross-platform comparison.
14. The method of claim 12, further comprising constructing an optimal content feature matrix by enumerating combinations of the highest-impact micro-pattern values and predicting engagement rates for each combination using a trained regression model.
15. The method of claim 12, wherein the significance threshold for recording an active micro-pattern is determined by the sample size, requiring a smaller correlation magnitude for detection when the sample size is large and a larger correlation magnitude when the sample size is small.
16. A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement: a learning acceleration framework that defines an analysis cycle duration and a human-equivalent analysis capacity computed from a learning rate multiplier and a capacity multiplier; a simultaneous A/B test runner configured to track a plurality of content variants, apply significance testing after a minimum impression threshold is met per variant, and extract learnings from all tests including inconclusive ones; a structural dimension analyzer configured to compute engagement correlations for at least

fifteen content formatting dimensions and maintain a registry of confirmed micro-patterns; a cross-platform synthesizer configured to classify confirmed micro-patterns as universal or platform-specific and construct a per-platform optimal content feature matrix; and a predictive model aggregator configured to combine timing, content, hashtag, hook, and format predictions into a composite engagement prediction with confidence bounds and feature attribution.

17. The computer-readable medium of claim 16, further storing instructions that implement a deadline pressure responder that reduces the scope of micro-pattern analysis and recommendation under increasing schedule pressure proportionally to the pressure magnitude.
18. The computer-readable medium of claim 16, wherein the A/B test runner stores test configurations, impression counts, engagement rates, and significance values in a persistent database and resumes in-progress tests after process restart.
19. The computer-readable medium of claim 16, further storing instructions that implement an optimal timing predictor that returns a recommended posting hour and day-of-week per platform, derived from a running map of per-platform per-hour engagement rate averages updated after each new content observation.
20. The computer-readable medium of claim 16, wherein the structural dimension analyzer aggregates micro-pattern records across a configurable rolling time window and removes patterns whose sample size has dropped below the minimum threshold due to data aging, ensuring the active micro-pattern registry reflects current rather than historical content behavior.

ABSTRACT

A HyperLearning engine for autonomous content optimization applies a 72x human-equivalent learning acceleration multiplier (derived from a 24x learning rate advantage and 3x capacity multiplier) to execute analysis cycles in 5 minutes that would require 6 hours of human analyst work. The engine simultaneously tests 30 content variants per cycle, declaring winners at 80% statistical significance with a minimum of 30 impressions per variant. A 15-dimensional micro-pattern detector computes Pearson correlations between structural content dimensions (character count, emoji density, hashtag position, hook structure, and 11 additional dimensions) and observed engagement rates, classifying confirmed patterns as universal (cross-platform) or platform-specific amplifiers. A composite predictive model combines timing, content, hashtag, hook, and format sub-models into engagement predictions with confidence bounds and factor attribution. A process-level in-memory cache prevents redundant database queries within analysis cycles. A deadline pressure responder narrows analysis scope under schedule pressure. Confirmed micro-patterns are applied as non-destructive formatting improvements to generated content before quality gate evaluation.